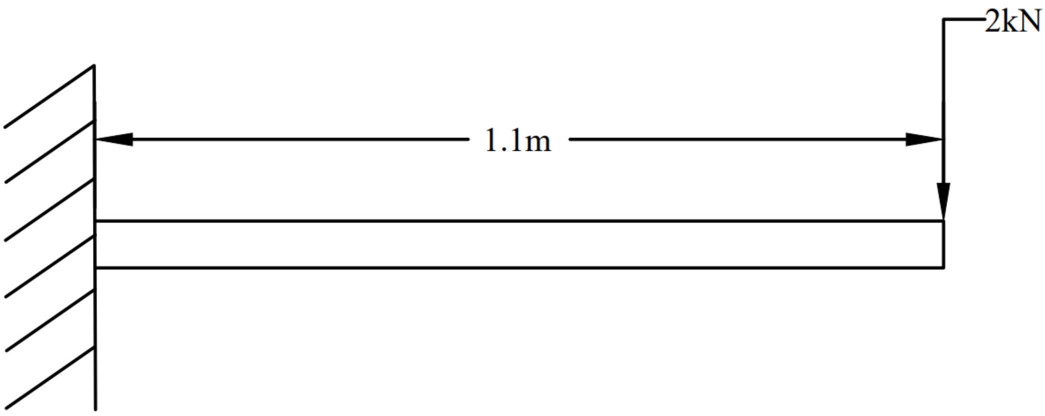
ENS161: Problem Set 6, Problem Set 7, and Quiz 7

Name of Group: *Truss Issues*
Section: *B6*

Part I.
For fill-in-the-blank questions, each blank is worth a point. For true-or-false questions, write only T or F. For identification questions, each desired item is worth a point.

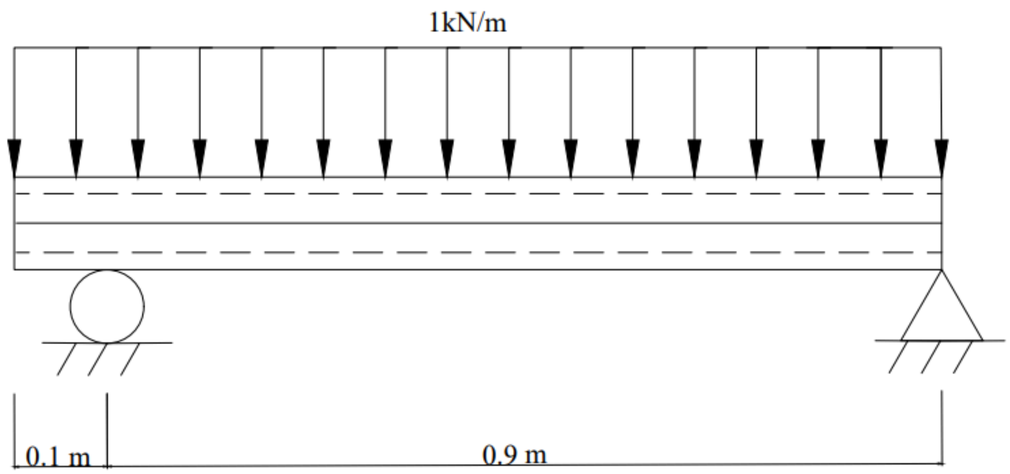
1. The ***free-body diagram*** of a rigid body represents it as having being isolated from its surroundings, replacing physical supports with forces and couple moments.
2. ***F*** *True or False*. An ideal truss does not have three-force members.
3. For 3D systems, the internal moment has two components: ***bending moment*** and ***torsional moment***, whose axes are tangent and normal to the cross section, respectively.
4. A body is said to be ***partially*** constrained if it experiences a fewer number of reactive forces than there are applicable equations of static equilibrium.
5. ***F*** True or False. A beam is subject to a load uniformly distributed throughout its length. The beam is supported at its left end by a pin, at its right end by a rocker, and at its midpoint by a roller. The beam is statically determinate.
6. For a simply supported thirteen-member plane truss to be considered simple, it must have ***8*** joints.
7. ***F*** True or False. A frictionless ball-and-socket joint resists translations and rotations.
8. A body is considered ***improperly*** constrained if all the reactive forces intersect at a common point or pass through a common axis, or if all reactive forces are parallel.
9. ***T*** True or False. A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another.
10. For internal loadings of 2D systems, the ***normal*** force and the ***shear*** force are normal and tangent to the cross section, respectively.

Scenario 1. A 1.1-m long cantilever beam is subject to a 2-kN weight at its free end.



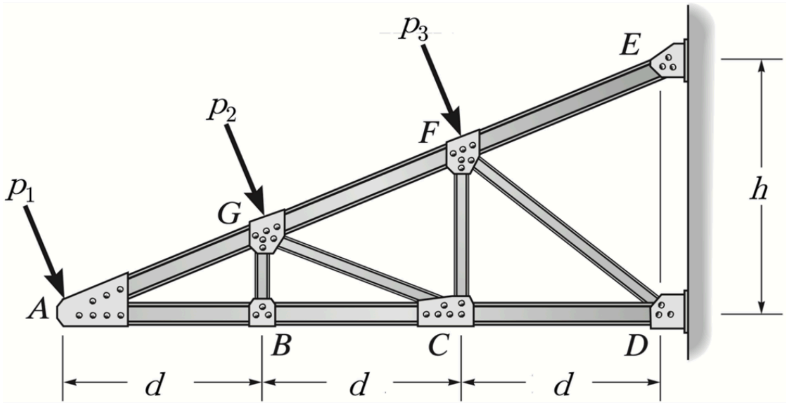
11. ***T*** True or False. This scenario is statistically determinate.
12. The support reactions have magnitudes of ***2*** kN and ***2.200*** kN-m.
13. The maximum bending moment developed is ***2.200*** kN-m in magnitude and occurs ***1.100*** m from the free end.
14. The maximum shear force developed is ***2*** kN-m in magnitude and occurs ***1.100*** m from the free end.
15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree ***0***.
16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree ***1***.

Scenario 2. A 1-m long beam carries a load uniformly distributed throughout its length. Assume a roller support 0.1 m from the left end and a pin support at the right end. The load has an intensity of 1 kN/m.



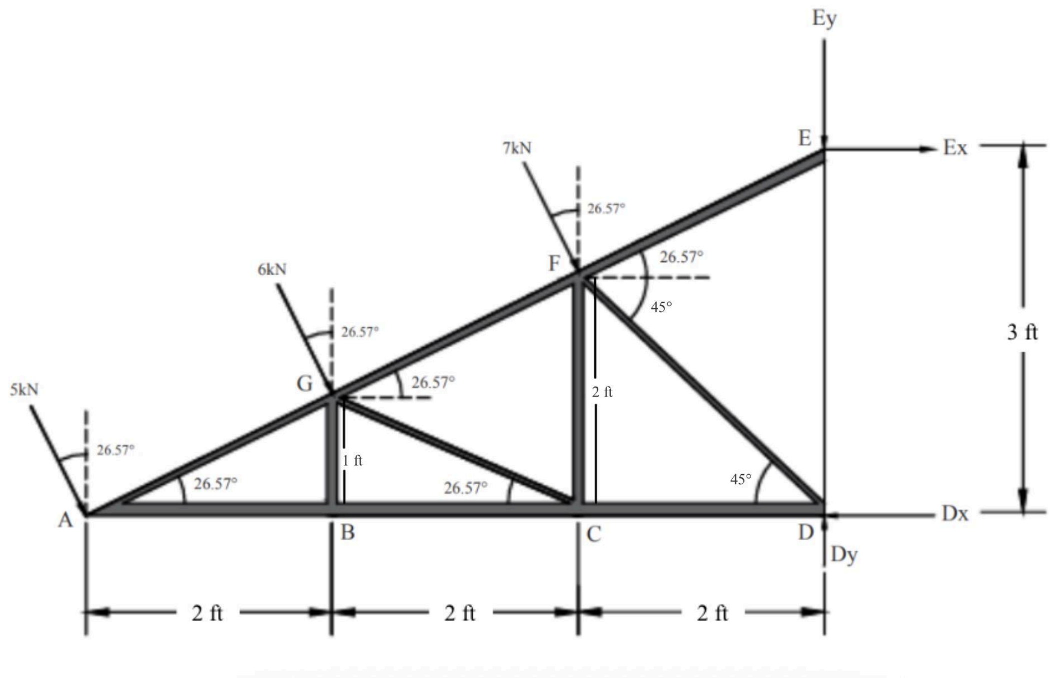
17. The roller support develops a force of ***0.556*** kN.
18. The pin support develops a force of ***0.444*** kN.
19. The bending moment developed at the right end is ***0.000*** kN-m.
20. The shear force developed just to the left of the pin is -***0.444*** kN.
21. The shear force developed just to the right of the pin is ***0.000*** kN.
22. ***F*** True or False. There is zero shear force developed at a point 0.9 from the right end.
23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree ***1*** .
24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **2**.

Part II.
Instructions: Where applicable, provide force diagrams and indicate the sense of member forces. Enclose final answers in boxes, with non-integer values rounded to three (3) decimal places.



Problem 1. Consider the truss subject to idealized loads as shown above. Assume that the members have negligible weights and that the joints behave as frictionless pins. The magnitudes of p1, p2, and p3 are 5 kN, 6 kN, and 7 kN, respectively. Set the dimensional parameters d and h to 2 ft and 3 ft, respectively.

Solution:



Given:

$h = 3\,ft$ $\qquad h_1 = \frac{3}{6}(4) = 2\,ft$ $\qquad h_2 = \frac{3}{6}(2) = 1\,ft$

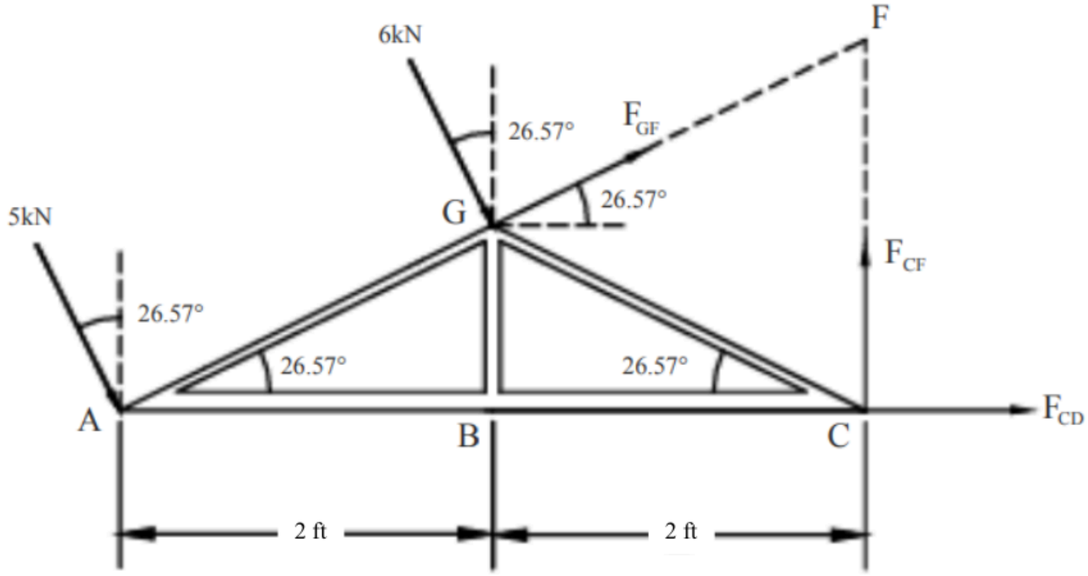
$d = 2\,ft$

$\theta = tan^{-1}(\frac{3}{6}) = 26.56505118°$

$\alpha = tan^{-1}(\frac{2}{2}) = 45°$

A.) (15 points) Determine the forces developed in members FG, CF, and CD. Use the method of sections, making sure to indicate the section cut and show the corresponding free-body diagram.

Cut the truss through FG, CF, and CD.



Taking moments about F: We take moments about point F to eliminate FG and CF so we can solve directly for F_{CD}.

$$\Sigma M_F: F_{CD}(2) + 6\sqrt{3} + (5\sqrt{3})2 = 0$$

Solving for F_{CD}: Substituting all external moment contributions gives the horizontal member force at CD.

$$\boxed{F_{CD} = -8\sqrt{5} = 17.889\ kN(C)}$$

Summing forces in x-direction to solve for F_{GF}

ΣF_x: $F_{GF}cos(26.57°) + (-17.889) + 6sin(26.57°) + 5sin(26.57°) = 0$

$\boxed{F_{GF} = 14.500\ kN\ (T)}$

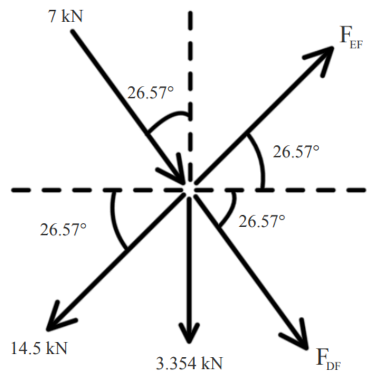
Summing forces in y-direction to solve F_{CF}

ΣF_y: $F_{CF} + 14.5sin(26.57°) = 6cos(26.57°) + 5cos(26.57°)$

$\boxed{F_{CF} = 3.354\ kN\ (T)}$

The problem is solved using the method of sections by cutting through members FG, CF, and CD, analyzing one side with equilibrium equations (∑M = 0, ∑Fx = 0, ∑Fy = 0) to find the member forces, and noting that when both support reactions lie on one side of the cut, the external forces on the cut section can be solved directly using moments.

B.) (10 points) Determine the forces developed in members DF and EF. Use the method of joints, making sure to show free-body diagrams at the joints considered. Hint: Use the computed forces in members FG, CF, and CD.



Summing forces in x-direction:

$$\Sigma F_x:\ F_{EF}cos(26.57°) + F_{DF}cos(45°) =- 7sin(26.57°) + 14cos(26.57°)$$

$$F_{EF}cos(26.57°) + F_{DF}cos(45°) = 9.8387$$

Summing forces in y-direction:

$$\Sigma F_y:\ F_{EF}sin(26.57°) + F_{DF}sin(45°) =- 7cos(26.57°) + 14sin(26.57°) + 3.354$$

$$F_{EF}cos(26.57°) + F_{DF}cos(45°) = 16.0997$$

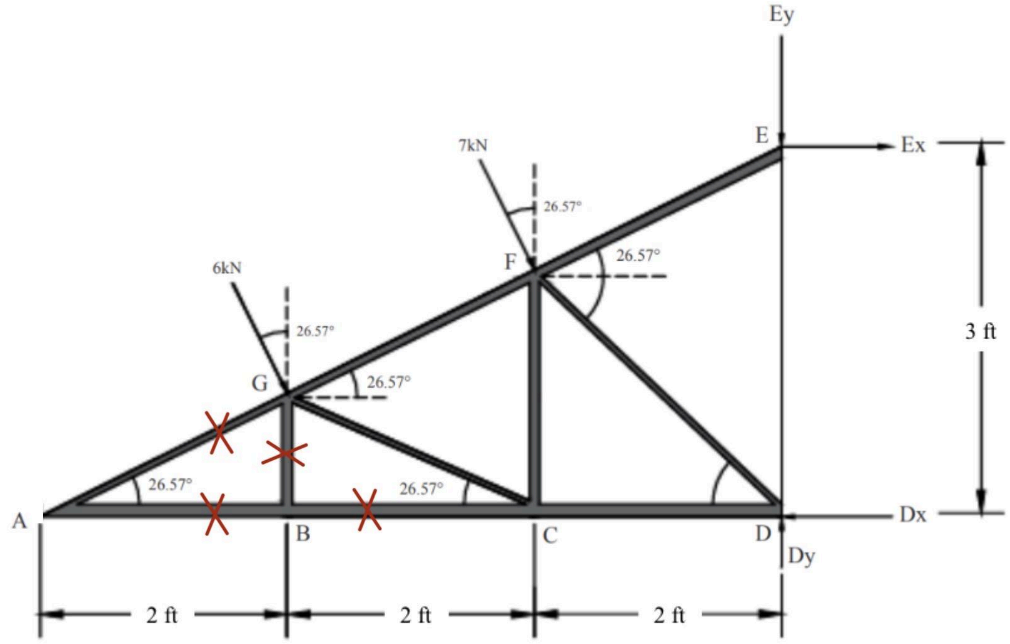
Solve the two equations simultaneously: Eliminate one unknown and solve for the other to obtain the magnitudes of F_{EF} and F_{DF}.

$$\boxed{\begin{aligned} F_{EF} &= 19.333\ kN\ (T) \\ F_{DF} &= 10.541\ kN\ (C) \end{aligned}}$$

C.) (5 points) If the truss is to be idealized as simply supported by a pin and a roller, at which joints must the pin and the roller be placed to maintain stability of the loaded truss? Justify your answer.

A pin at joint A and a roller at joint D. A pin at joint A provides the needed horizontal and vertical reactions, while a roller at joint D provides only a vertical reaction, so the truss remains statically determinate and free to expand. We cannot reverse them because the wall at D cannot supply a horizontal reaction. Modeling it as a pin would over-constrain the truss. In addition, the inclined loads create significant horizontal components, so the structure must have one support capable of resisting both x and y directions, which is provided by the pin at A. The roller at D must allow horizontal movement and therefore cannot contribute a horizontal reaction, keeping the truss compatible and preventing redundant constraints. This configuration also ensures rotational stability, because placing a horizontal reaction at D would create an improper force couple and over-restrict the structure.

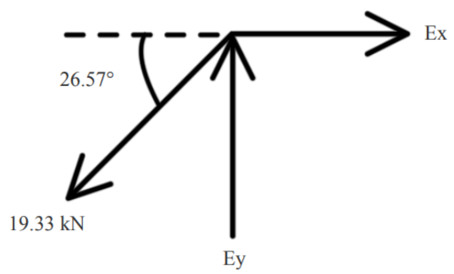
D.) (4 points) A contingency is considered wherein joint A is effectively unloaded. Identify zero-force members, if there be any, for this case.



$$\boxed{F_{AG} = F_{AB} = F_{BC} = F_{GB} = 0}$$

E.) (6 points) Determine the forces developed in the supports at E and at D. Hint: Use the computed forces in members CD, DF, and EF.

Methods of Joints at E to solve for reactions Ex and Ey



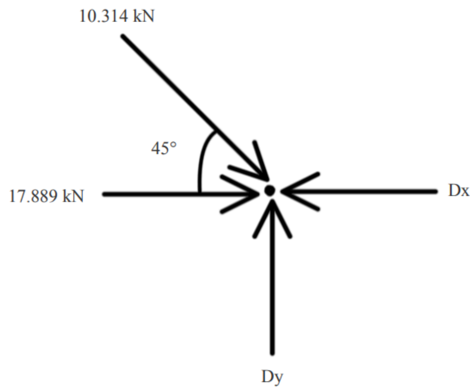
Summing forces in x-direction:

$$\boxed{\Sigma F_x:\ E_x = 17.292\ kN}$$

Summing forces in y-direction:

$$\boxed{\Sigma F_y:\ E_y = 8.646\ kN}$$

Methods of Joints at D to solve for reactions Dx and Dy



Summing forces in x-direction:

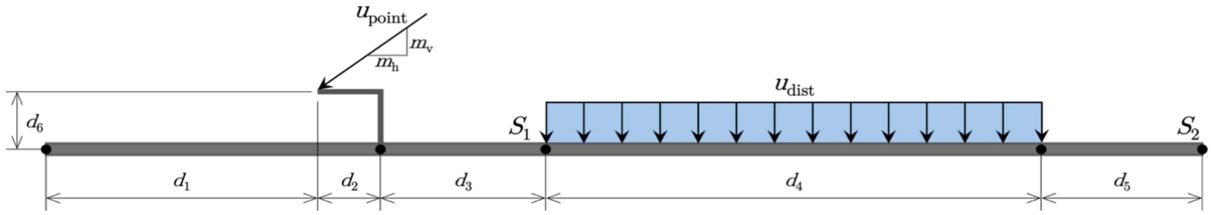
$$\Sigma F_x : D_x = 10.541cos(45°) + 17.889$$

$$\boxed{D_x = 25.342\ kN}$$

Summing forces in y-direction:

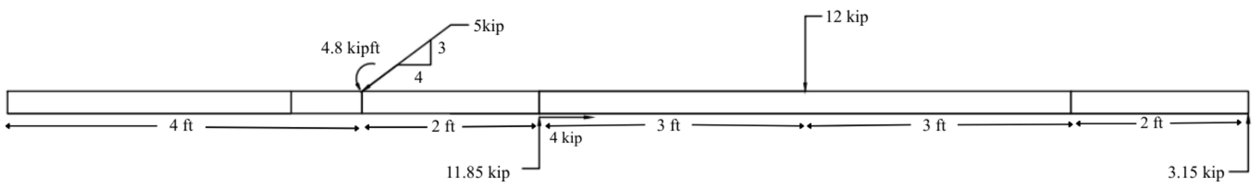
$$\Sigma F_y : D_y = 10.541sin(45°)$$

$$\boxed{D_y = 7.454\ kN}$$



Problem 2. A custom beam design places supports at S1 and at S2, in anticipation of a concentrated load of upoint kips and a uniformly distributed load of intensity udist kip/ft. Your main task is to analyze how the internal loadings induced vary as a function of x, where x is the distance from the left end of the beam. Consider the case when upoint is 5 kips and udist is 2 kip/ft. Set the respective values of d1, d2, d3, d4, d5, d6 to 3.2, 0.8, 2.0, 6.0, 2.0, and 0.6 ft, and those of mv and mh to 3 and 4. For static determinacy, assume a pin at S1 and a roller at S2.

a. (6 points) Determine the support reactions.
b. (1 point) Determine the normal force at d1 + d2 from the left end.
c. (2 points) Determine the shear force at d1 + d2 from the left end.
d. (3 points) Determine the bending moment at d1 + d2 from the left end.
e. (8 points) Determine the shear force as a function of x.
f. (8 points) Determine the bending moment as a function of x.
g. (6 points) Plot the shear force diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.
h. (6 points) Plot the bending moment diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.



Transferring the applied force to the beam axis at x = 4.0 ft.

$$\Sigma M_C:\ 3(0.8)\ +\ 4(0.6)$$
$$= 4.8\ kip \bullet ft$$

Equivalent Uniform Distributed Load force and its centroid location:

$$F_R = 2(6)\ =\ 12\ kip$$
$$d\ =\ \frac{6}{2}\ =\ 3\ ft$$

A. Support Reactions

$$\Sigma M_D:\ 4.8\ +\ 3(2)\ -\ 12(3)\ +\ R_y\,(8)\ =\ 0$$
$$\boxed{R_y\ =\ 3.15\ kips}$$

$$\Sigma F_x:\ 3\ +\ 12\ = D_y\ +\ 3.15$$
$$\boxed{D_y\ =\ 11.85\ kips}$$

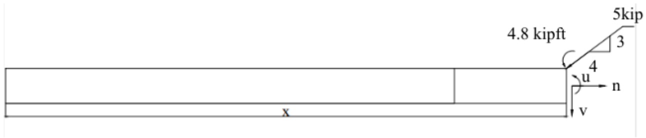
$$\Sigma F_x:\ \boxed{D_x\ =\ 4\ kips}$$

B-D. Normal force, Shear force, Bending moment at d1 + d2 from the left end

$x = 4^{-}$	$x = 4^{+}$
$n = 0\ kip$	$n = 4\ kip$
$v = 0\ kip$	$v = -3\ kip$
$\mu = 0\ kip$	$\mu = -4\ kip \bullet ft$

E-F. Shear force and Bending moment as a function of x

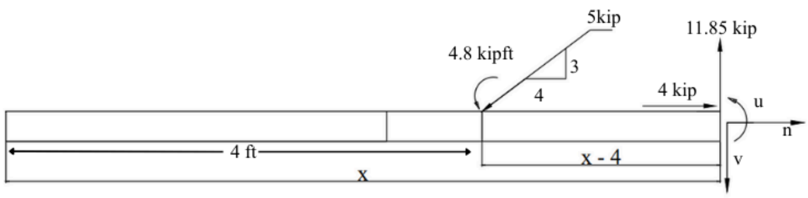
@ $x < 4$



$n = 0\ kip$
$v = 0\ kip$
$\mu = 0\ kip$

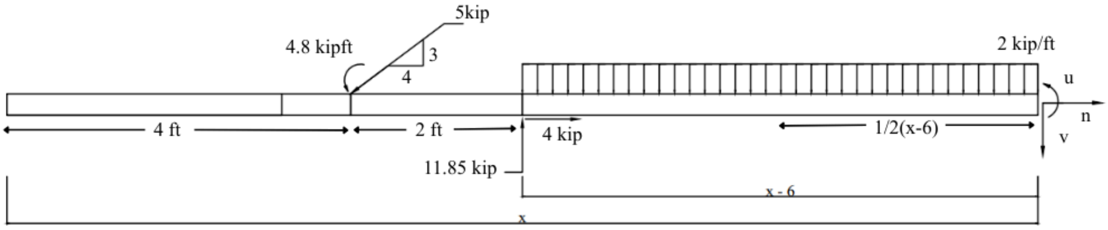
@ $4 \leq x < 6$

$n = 4\ kip$
$v = -3\ kip$
$\mu = -4.8(3)(x - 4)\ kip \bullet ft$
$= 7.2 - 3x$



@ $6 \leq x < 12$

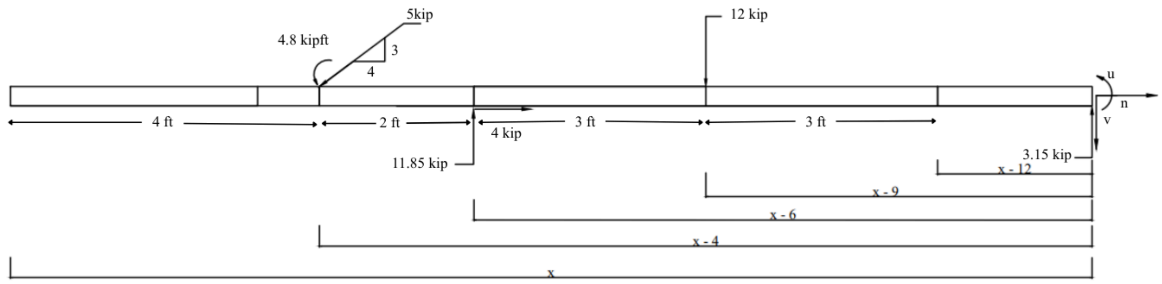
$n = 0\ kip$
$v = -3 - 2(x - 6) + 11.85\ kip$
$= 20.85 - 2x\ kip$
$\mu = -4.8 - (3)(x - 4) + 11.85(x - 6) - 2(x - 6)(x - 6)(\frac{1}{2})\ kip \bullet ft$
$= -x^2 + 20.85x - 99.9\ kip \bullet ft$



@ $12 \leq x < 14$

$n = 0\ kip$
$v = -3 + 11.85 - 12\ kip$
$= -3.15\ kip$
$\mu = -4.8 - (3)(x - 4) + 11.85(x - 6) - 12(x - 9)\ kip \bullet ft$

$$= 44.1 - 3.15x\ kip \bullet ft$$

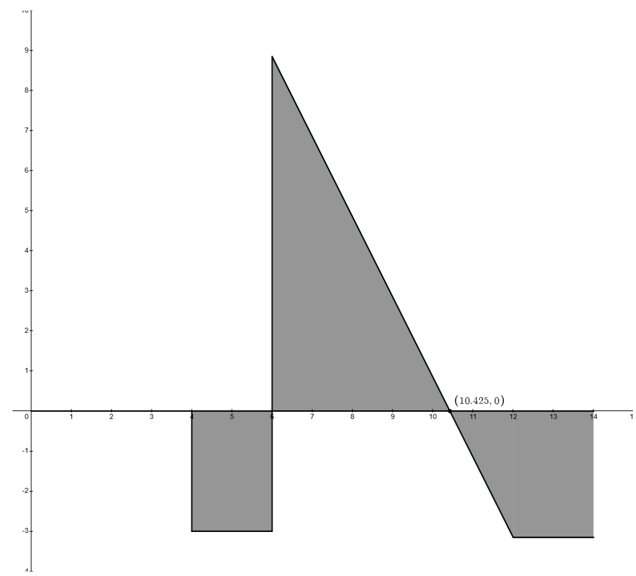


Summary:

G. Shear Force V(x) in kips

v(x)

$$= \begin{cases} 0 & x < 4 \\ -3 & 4 \leq x < 6 \\ 20.85 - 2\,x & 6 \leq x < 12 \\ -3.15 & 12 \leq x < 14 \end{cases}$$



H. Bending Moment M(x) in kip-ft.

$$\mu(x) = \begin{cases} 0 & x < 4 \\ 7.2 - 3\,x & 4 \leq x < 6 \\ -x^2 + 20.85\,x - 99.9 & 6 \leq x < 12 \\ 44.1 - 3.15\,x & 12 \leq x < 14 \end{cases}$$